

Adaptive Learning in Extended Reality: A Survey on multimodal Interaction and AI-driven Personalization

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ABSTRACT

Extended Reality (XR) is transforming the way users engage in learning and training through immersive, interactive environments. This survey analyzes adaptive XR learning systems, focusing on AI-driven personalization and multimodal interaction. We examine how systems use diverse natural signals such as gaze, speech, gestures to model user states and dynamically adapt feedback in real-time. Unlike prior reviews that treat XR-based systems, multimodal integration and adaptive learning independently, this work provides a detailed synthesis of their convergence by reviewing recent developments in XR learning environments, highlighting the use of large language models (LLMs) for context-aware adaptation and guidance generation. Key platforms, including SMARTe-VR, XaiR and SIGMA, are examined alongside multimodal benchmarks like Ego4D, HOT3D, and HoloAssist. By identifying architectural trends and adaptation strategies, we highlight ongoing challenges such as data fusion, user modeling and data privacy concerns. This survey outlines future directions toward scalable, highly personalized XR learning systems powered by AI and rich sensor data.

Index Terms: Extended Reality, Large Language Models, Adaptive Learning, Multimodal Data, AI-Driven Personalization.

1 INTRODUCTION

Learning and training are being transformed by major technological advances that aim to improve skill acquisition and knowledge retention. Extended Reality (XR), encompassing the virtual continuum of Virtual Reality (VR), Augmented Reality (AR) and Mixed Reality (MR) [28], offers immersive, interactive and personalized digital learning environments, enabling the simulation of complex real-world scenarios and the creation of diverse augmented experiences [27]. Applications span manufacturing, medical training, soft skills development, supporting embodied training and contextualized feedback surpassing traditional approach [25, 5].

However, as education and training methods evolve, there is a clear need for dynamic and personalized XR systems that adapt learning in real time based on user performance, cognitive or emotional states [34, 39, 30]. In addition, modern XR technologies are increasingly leveraging artificial intelligence (AI), especially large language models (LLMs) in conjunction with multimodal sensor data, focusing on developing generalizable systems capable of generating dynamic individualized guidance that moves beyond static content delivery.

XR systems can capture rich multimodal signals such as gaze, gesture, speech, physiological signals and spatial context within 3D environments [19, 33], providing deep insight into user engagement and cognitive load during training [41, 30]. For example, an XRXL system [3] has boosted large-scale educational engagement and industry deployments have revealed significant ad-

vancements in employee retention and training outcomes through immersive, data-driven feedback [1]. However, effective real-time synthesis and interpretation of multimodal data impose a significant research challenge, highlighting the need for advanced data fusion, context awareness and user modeling, whilst ensuring usability, transparency, and ethical management. This often requires implementation of cutting-edge AI technologies, including multimodal LLMs [24, 40, 8, 9].

This paper presents an overview of adaptive learning in XR, with a focus on multimodal interaction and AI-driven personalization, highlighting key trends and challenges in their integration.

2 BACKGROUND ON KEY CONCEPTS

Extended Reality: Definitions and Components. XR technologies enhance or alter users' perceptions of reality by partially or fully immersing them in virtual content, blending digital and physical worlds. Leveraging high-fidelity graphics, low-latency tracking, and 6DoF motion capture, VR systems replace the real world, enabling immersive interaction across a wide variety of applications. In contrast, AR systems are grounded in the real world and enhance users' perception of their physical environment by overlaying digital content via real-time camera and sensor input. They are widely used in navigation, education, and collaboration. MR systems, while encompassing AR, often require more advanced spatial mapping, anchor tracking, and real-time data fusion to enable seamless interaction between the real and virtual worlds. A technical report by Accenture [1], along with a comprehensive review of XR applications [19], highlight the effectiveness of these systems for collaborative and procedural tasks.

Extended Reality for Learning and Training: Beyond Traditional Education. In learning and training, XR facilitates experiential learning, increases engagement and knowledge retention by providing immersive contexts often impractical or costly to replicate in reality. Its scope extends beyond traditional education, providing valuable for academic training, skill acquisition, workforce or personal development [37, 22, 32].

Multimodal Data Collection Nuances in VR, AR and MR. The nature of multimodal data collection and interpretation varies significantly across XR subdomains, each of which poses unique challenges. In case of VR, data is captured in synthetic, controlled environments, ensuring high-fidelity tracking but lacking real-world variability; AR requires real-time interpretation of the physical world, where challenges such as occlusion, lighting changes or generated content alignment demand strong spatial and context awareness; MR, by combining virtual and physical worlds, involves continuous tracking of spatial anchors and interaction points, resulting in more complex data fusion [28]. These differences shape XR adaptive mechanisms and multimodal data fusion strategies [42].

Multimodal Interaction and Integration in XR. As data collection capabilities evolve, modern XR environments have begun to integrate diverse interaction modalities, providing deeper understanding of user behavior. Tab. 1, adapted from Kourtesis [19], summarizes the primary sensing modalities used in adaptive XR systems and their typical applications. In addition, galvanic skin response (GSR) provides valuable information on user stress level, particu-

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Table 1: Common XR sensing modalities and their applications. Adapted and extended from [19].

Modality	Description
Eye/Gaze Tracking	Measures visual attention or gaze direction; supports adaptive guidance and engagement monitoring
Hand/Gesture Tracking	Detects hand motion and gestures for direct interaction; enables natural control and virtual manipulation
Facial Tracking	Captures facial features and expressions; supports emotion-aware systems and social cues
Full-Body Tracking	Tracks body pose and joint position; useful for posture-aware interaction and motion-based instruction
EEG	Records brain activity; supports real-time estimation of mental state and cognitive load

larly useful in emotionally adaptive interfaces. Voice and speech recognition enables hands-free interaction and verbal guidance systems [18]. XR systems also often employ scene understanding approaches, such as spatial mapping, semantic scene analysis, which complement user-input interpretation but rely on different sensor setups [42, 35]. The rise of multimodal sensors has increased the demand for models capable of interpreting multiple data streams in real-time. This has contributed to the rising popularity of multimodal LLMs (mLLMs), which enable advanced interpretation of complex signals. For example, XR-Objects [13] uses mLLMs for object recognition and semantic understanding automation in MR. The shift toward mLLM-driven architectures marks a major step beyond traditional rule-based approaches.

Datasets and Benchmarks for Multimodal Adaptive XR. Integrating these diverse modalities provides a comprehensive, real-time view of the user, enabling more effective personalization. Key multimodal datasets such as Ego4D [14], HOT3D [2], and HoloAssist [41] provide standardized benchmarks for training and evaluating models that interpret complex human behaviors in immersive environments. A recent survey by Li and Lu [21] highlights the growing importance of such benchmarks for mLLMs, underscoring their importance for advanced human behavior analysis.

Principles of Adaptive Learning and AI-driven Personalization in XR. Adaptive learning systems dynamically adjust guidance based on real-time user performance, preference, and cognitive state to optimize learning, foster engagement, and skill acquisition. These systems operate through continuous feedback loops involving user state modeling, adaptive strategies (e.g., adjusting difficulty or feedback types), real-time content adaptation and personalized delivery through visual and vocal cues, virtual tutors, and dynamic interfaces [43, 13, 20]. According to Yichi Zhang et al. [43], task guiding systems have evolved from early rule-based approaches to perception-enabled, task-specific solutions. Recent research has primarily focused on developing components for single-domain systems, including environment understanding, user behavior, and cognitive state analysis [16, 17, 23].

3 METHODOLOGY FOR PAPER SELECTION AND ANALYSIS

A targeted search was conducted in *IEEE Xplore*, *ACM Digital Library* and *ArXiv* using keywords including “adaptive learning in XR”, “multimodal interaction”, “AI in XR education” and related terms. Papers from 2020–2025, published in top-tier conferences (e.g., IEEE VR, IEEE ISMAR, ACM CHI, SIGGRAPH, ACM UIST) were prioritized to ensure quality and relevance. Studies were selected based on whether they addressed adaptive learning mechanisms in XR, use of multimodal interactions and strategies for sensor data integration. From an initial pool of over 40 articles, six studies were selected for in-depth analysis based on their shared focus on adaptive XR learning, particularly through multimodal interaction and AI-driven personalization.

4 REQUIREMENTS AND CHALLENGES FOR ADAPTIVE XR SYSTEMS

Adaptive XR training systems must address several requirements, shaped by the immersive and interactive nature of XR:

a) Human-Centered Adaptivity: Systems should adapt content, task guidance and interfaces to user abilities, context and goals, supporting personalized profiles and real-time adjustments for collaborative and individual scenarios, possibly via AI-based continuous user state modeling and feedback adjustment [17, 7, 38, 20].

b) Multimodal Interaction and Data Integration: Modern XR leverages diverse multimodal input, requiring robust fusion and interpretation of high-dimensional data, ensuring seamless, low-latency and reliable interaction [42, 19, 8]. Cutting-edge deep learning and multimodal fusion can be employed to effectively combine and interpret sensor data streams.

c) Context and Spatial Awareness: XR systems require continuous sensing and interpretation of spatial context for precise alignment between physical and virtual objects, avoid occlusions and maintain comfortable user experiences, traditionally achieved through SLAM (Simultaneous Localization and Mapping), spatial mesh reconstruction and real-time object tracking [11, 35, 20, 8].

d) Transparency and Security: As LLM-based agents process highly sensitive behavioral data [17], privacy protection and transparent operation are essential, with explainable AI contributing to the interpretability of data processing and adaptation mechanisms.

e) Scalability and Real-world Deployment: XR applications must scale across devices, domains and user profiles while remaining robust amid uncertainty and integration challenges [41, 7, 6]. According to Liubogoshchev et al. [26] scalable XR deployment can be supported by cloud-based architectures combined with on-device processing, modular system design, and hardware-specific optimization to balance performance, latency and cost.

f) Proactive Guidance and Human-AI Collaboration: Valuable insights from HoloAssist [41] have revealed new criteria for efficient assistance in XR settings: fine-grained mistakes detection and proactive instructor intervention; context and spatial-aware task guidance; multimodal sensor fusion for user intent and action prediction. Such capabilities can be implemented through predictive AI trained on expert demonstrations, enabling the system to intervene before critical errors occur.

These findings suggest that next-generation adaptive XR systems should move beyond reactive, query-driven support and instead adopt proactive, context- and spatial-sensitive real-time assistance. This raises new challenges for AI models in prediction, multimodal understanding, and real-time intervention in complex physical environments.

5 SYSTEM DESIGN AND COMPONENTS

Adaptive XR systems integrate hardware, software and AI components into an overall system design to insure adaptive, personalized user experiences. Recent research underscores several system-level architectural trends: embedded multimodal sensors in head-mounted devices enable efficient capture and analysis of natural user interactions; LLMs/mLLMs integration allows for real-time personalized content, guidance and feedback, especially in dynamic learning and collaborative scenarios; spatial computing is critical for AR/MR, relying on SLAM, spatial anchors and environmental sensing to ensure accurate alignment with the physical world; platforms such as Microsoft’s Platform for Situated Intelligence [6] provide infrastructure for streaming, synchronizing and processing multimodal sensor data, supporting rapid prototyping of adaptive, explainable XR agents.

Fig. 1 illustrates the overall system architecture and workflow of an LLM-driven adaptive XR training system. The diagram is divided into three main phases. Firstly, the system records various signals, for example, egocentric video, speech narration, gaze

tracking, spatial mesh, while expert performs a detailed demonstration. Then, a multimodal LLM analyzes the captured signals and generates detailed instructions. The third module is responsible for providing guidance to a user based on generated instructions and contextual information.

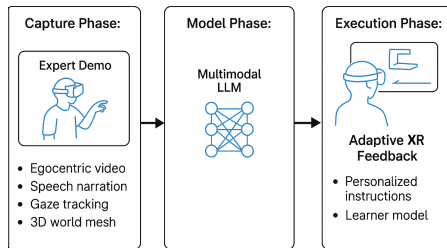


Figure 1: System design of an LLM-driven adaptive XR training system.

This design enables adaptive guidance and coaching in XR environments by linking expert demonstrations, advanced AI processing, and real-time, learner-specific feedback within immersive training scenarios.

6 COMPARATIVE ANALYSIS OF KEY STUDIES ON ADAPTIVE XR LEARNING SYSTEMS

6.1 In-depth Analysis of Key Studies

a) *Exploring Large Language Model-Driven Agents for Environment-Aware Spatial Interactions and Conversations in Virtual Reality Role-Play Scenarios [23]*: This work introduces a system that utilizes LLMs to power conversational AI agents that engage in spatial-aware interactions within VR role-play scenarios. The agent dynamically adapts response based on user query and spatial context. The key contribution is to demonstrate the power of LLMs in generating dynamic, personalized conversational experiences that respond to linguistic and spatial cues. However, the primary focus is not explicit user modeling but local dialogue adaptation.

b) *XaiR: An XR Platform that Integrates Large Language Models with the Physical World [35]*: XaiR proposes an XR platform integrating LLMs with real-world sensor data to create intelligent, adaptive, context-aware XR experiences. The platform uses diverse multimodal sensory inputs like camera feeds and spatial sensors, combined with natural language interaction, to dynamically adjust content, whilst LLM acts as a central reasoning engine. XaiR's key contribution is an architectural framework that bridges LLMs and physical XR interactions. However, the paper lacks specific adaptive learning mechanisms or empirical validation of learning outcomes, with real-time performance and robustness in dynamic physical environments.

c) *Autonomous Workflow for Multimodal Fine-Grained Training Assistants Towards Mixed Reality [29]*: This work introduces an autonomous workflow for building fine-grained training assistants in MR, providing context-aware guidance during procedural tasks. Leveraging multimodal inputs (hand gestures, eye gaze, spatial tracking and voice commands), the system detects user errors and provides timely instructions. While this work shows a structured approach to integrating multimodal data and vision-language models, the exploration of personalization, long-term learning retention and task scalability is limited.

d) *SIGMA: An Open-Source Interactive System for Mixed-Reality Task Assistance Research [7]*: SIGMA is an open source platform for mixed reality task assistance research. It enables dynamic instructions, context-aware overlays and personalized feedback based on user actions and task progress, using multimodal in-

Table 2: Summary of Adaptive XR Systems.

System	LLM/AI Adaptivity	Multimodal Inputs, User Modeling
Role-Play VR	Conversational LLM; context-aware dialogue adaptation	Voice and spatial cues; implicit modeling for role-play learning
XaiR	LLM as central reasoning engine; real-world context adaptation	Camera, spatial data, voice; basic state tracking for XR environment intelligence
Workflow MR	ML/LLM-based error detection; procedural task adaptation	Gesture, gaze, voice, spatial input; action-level modeling for training assistance
SIGMA	Modular AI logic; dynamic task instructions and overlays	Gesture, gaze, position, voice; developer-defined modeling for task assistance
Explainable XR	LLM-based analytics; offline analysis without in-situ adaptation	Gaze, movement, verbal actions; post-hoc behavior modeling and interpretation
SMARTe-VR	AI-based inference of engagement and frustration; real-time adaptive alerts	Biometrics, gaze, navigation logs; cognitive/emotional modeling in VR learning

puts like hand gestures, spatial tracking, eye gaze, and voice commands to understand user intent. SIGMA provides the core infrastructure to test AI-driven task understanding and adaptive guidance. Its key strength is lowering the barrier for MR task assistance research, although it lacks details on adaptive learning outcomes or advanced personalization, as these would be the subject of research conducted using the platform.

e) *Explainable XR: Understanding User Behaviors of XR Environments using LLM-assisted Analytics Framework [17]*: Explainable XR introduces an LLM-assisted analytics framework to interpret complex user behaviors in XR. Although not an adaptive system itself, it provides foundational insights, derived from multimodal data, to support personalization based on a deep understanding of user actions, intentions, and cognitive states. LLM synthesizes these data into explainable behavioral models, informing adaptive decisions and increasing system transparency. A key limitation is the lack of real-time user behavior analysis and integrated adaptation mechanisms.

f) *SMARTe-VR: Student Monitoring and Adaptive Response Technology for e-Learning in Virtual Reality [12]*: SMARTe-VR is a platform for immersive VR e-learning focused on student state monitoring and adaptive response. It infers engagement, frustration and performance, using biometrics, gaze tracking, interaction logs, and spatial navigation, which enables real-time user modeling and state inference via AI/ML. This method addresses the key challenge of collecting and analyzing rich data for effective personalization in VR. However, SMARTe-VR does not cover adaptation strategies or automated content generation, serving as an enabling technology rather than a complete adaptive tutoring system.

A detailed comparison table of the key studies can be found in the supplemental materials.

6.2 Common Trends

Recent analysis demonstrates a drastic increase in publications at the intersection of AI (LLMs), XR and learning. According to Dai and Nuder [10], the annual number of publications on AI-powered XR learning increased from 38 in 2020 to 94 in 2023, with 80 al-

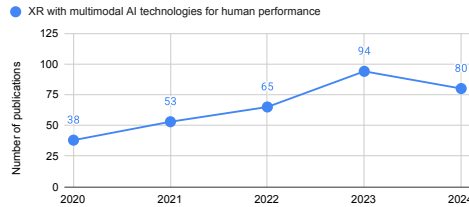


Figure 2: Publications trend by year (2020–mid 2024) adapted from Dai and Nuder (2024) [36].

ready published by mid-2024 Fig. 2, likely driven by the rise of LLMs and multimodal capabilities of XR. In particular, Tang et al. [36] highlight a rapid growth of multimodal interaction and data fusion in XR learning systems, powered by LLMs, indicating a shift toward data-driven, adaptive learning environments. Other recent reviews [15, 31] and related works along with the key studies, point to the same tendency and identify several common trends:

a) Rapid adoption of LLMs: Increasingly, studies incorporate LLMs/mLLMs to interpret and synthesize user input across modalities, indicating a shift from rule-based to data-driven adaptive systems and AI-driven personalization [24].

b) Central role of multimodal interaction: Studies emphasize diverse input signals (gaze, voice, gestures, spatial, biometrics) which has become foundational for effective user experience in XR [19].

c) Conversational, context-aware AI agents: There is a clear rise of AI agents in XR capable of natural language interaction and context-aware behavior for more engaging adaptive experiences [23].

d) User state modeling: While explicit user modeling is not yet common, advanced user cognitive state inference is an emerging and critical trend since accurate user modeling, especially through multimodal data fusion and LLMs integration, is the foundation for the next-generation adaptive, human-centric XR strategies [30, 17].

6.3 Diverse Approaches and Methodologies

Despite common trends, both the key papers analyzed and the broader cutting-edge research in adaptive XR training demonstrates diverse methodologies and focus areas. Some works offer complete system architectures [35, 4], while others provide open-source platforms with foundational infrastructure for future research [6, 7]. A significant scope of papers focus on applying novel AI models for specific adaptive tasks like content generation or user state inference [16]. Moreover, some works showcase analytical frameworks crucial for understanding user behavior and interaction [17, 30, 41]. While majority are adaptive, the type of adaptation varies from conversational guidance to fine-grained procedural assistance and personalized content creation [37, 29, 23].

7 CHALLENGES AND OPEN ISSUES

Despite significant progress in adaptive XR learning systems, several key challenges and open issues can be identified:

a) Multimodal data collection and fusion: Collecting and synchronizing diverse high-dimensional sensor data remains challenging due to the frequent data misalignment, environmental noise and large data volumes limiting storage and real-time processing.

b) User profile modeling and adaptive algorithm design: Accurately modeling of cognitive and emotional states in real-time often relies on simplified assumptions. Models struggle to generalize across users, face “cold start” issues with new learners and suffer from AI’s low explainability.

c) User experience in XR environment: Complexity, overabundance or irrelevance of XR guidance can significantly affect users’

comfort and engagement, impacting the effectiveness of adaptive learning.

d) Scalability and deployment constraints: Deploying XR systems beyond prototypes is impeded by high computational demands, hardware requirements, and integration barriers limiting widespread adoption of adaptive XR learning systems.

e) Ethical considerations: Extensive personal data collection in XR raises significant privacy and security concerns.

Beyond these technical and ethical challenges, there remains a broader conceptual and practical gap: *Recent studies confirm that adaptive learning is a recognized direction, but remains far from being a mainstream reality in practical XR systems.* While many frameworks incorporate elements of adaptivity (e.g., context-aware feedback, multimodal sensing), fully realized adaptive learning environments, capable of continuously and intelligently adjusting to learner states, are still rare. This highlights an urgent need for integrative, scalable, and user-centered approaches that bridge foundational research with deployable solutions.

8 FUTURE DIRECTIONS

Considering the identified challenges and the advancements in current research, several key future directions will shape the next generation of adaptive XR learning environments:

a) Advanced AI for better dynamic adaptation and content generation: Stronger AI models might potentially support real-time content generation, spatial computing and semantic scene understanding in XR. Emphasis should shift toward proactive models for more responsive adaptation.

b) Multimodal sensing automation: Future systems should develop AI-driven sensor fusion to autonomously extract complex user states from raw multimodal data.

c) Developing standardizes frameworks for data integration and interpretation: Unified data models, APIs, datasets and benchmarks could foster interoperability across platforms, faster integration and research.

d) Long-term efficacy studies: Future research must prioritize in-depth, long-term empirical studies to validate adaptive XR’s impact on learning outcomes.

9 CONCLUSION

This survey examined the rapidly expanding field of adaptive learning in XR, highlighting the synergistic role of multimodal interaction and advanced AI. Recent studies reveal a clear trend toward learning experiences that are more personalized, engaging, and effective, with LLMs/mLLMs enabling richer user modeling, adaptive content generation, and natural multimodal interaction. Nevertheless, major challenges remain, including the complexity of multimodal fusion, computational demands, and privacy concerns, which may limit near-term deployments to partial or superficial integrations. This survey is limited by its focus on studies combining XR, AI, and input modalities, narrowing coverage relative to domain-specific reviews. As research advances, these approaches could transform education and training and extend to fields such as manufacturing and healthcare, enhancing personalization, workforce readiness, and social impact.

SUPPLEMENTAL MATERIALS

A detailed comparison table of key studies is available on OSF at <https://doi.org/10.17605/OSF.IO/DP975>.

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